

## RESEARCH STATEMENT

My main research interests lie in AI for Science, especially its applications in materials science and fluid simulation.

## EDUCATION

**Department of Mathematics, Hong Kong University of Science and Technology**  
 Hong Kong SAR, China

*Incoming Ph.D. Student in Mathematics* 2026.09 –  
 • Advisor: Prof. Xiang Yang.

**School of Data Science, The Chinese University of Hong Kong, Shenzhen** Shenzhen, China

*B.Sc. in Data Science and Big Data Technology* 2022.09 – 2026.07  
 • CGPA: 3.89/4.00; Major Rank: 7/150.  
 • Dean's List: AY2022–2023, AY2023–2024, AY2024–2025.  
 • Academic Performance Scholarship, Class B: AY2023–2024, AY2024–2025.

**University of California, Berkeley** Berkeley, CA, USA  
*Berkeley Global Access Program, Visiting Student* 2024.08 – 2024.12  
 • GPA: 4.00.

## EXPERIENCES

**MPM-Based Fluid Simulation Algorithms for Mobile Physics Engines** | Supervised by Dr. Zhihao Shen (Huawei) and Prof. Xiang Yang (HKUST) 2026.01 – Present  
 • Studied fluid simulation algorithms, especially the Material Point Method (MPM).  
 • Reproduced MLS-MPM using taichi\_mpm and developed a dam-break simulation demo to validate the algorithm pipeline.  
 • Reproduced IQ-MPM with ZIRAN and integrated its C++ simulation backend into a HarmonyOS application through DevEco Studio, with ongoing work on mobile-side runtime adaptation and performance optimization.

**Deep Generative Models for Boltzmann Distribution Sampling** | Supervised by Prof. Kai Zhou, CUHK-SZ 2024.08 – 2025.08  
 • Studied the theoretical foundations of score-based and flow-based generative models, with emphasis on their connections to statistical physics and Boltzmann distribution sampling.  
 • Analyzed *Energy-Based Diffusion Generator for Efficient Sampling of Boltzmann Distributions* and reproduced its main numerical experiments.  
 • Explored a stochastic-interpolation-based modification to the original sampling procedure.

**Learning-Augmented Wind Power Optimization** | Supervised by Dr. Jingyi Zhao, Shenzhen Research Institute of Big Data 2024.12 – 2025.06  
 • Studied Learning to Optimize (L2O) methods and mixed-integer nonlinear programming (MINLP) formulations for data-driven decision-making under operational constraints.  
 • Implemented a *predict-then-optimize* pipeline for a real-world wind power optimization problem in Shanxi Province, integrating MoE-based forecasting with MINLP-based constrained optimization.

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|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| PROJECTS    | <b>Comparing Traditional Methods and Diffusion Models for Hybrid Image and Visual Anagram Generation Tasks</b>   2025 Summer                                                                                                                                                                                                                                                                            |  |
|             | <ul style="list-style-type: none"> <li>Implemented hybrid image and visual anagram generation using diffusion model and traditional methods such as image pyramids.</li> <li>Systematically compared the performance using multiple metrics such as visual quality and CLIP similarity score.</li> </ul>                                                                                                |  |
|             | <b>Building Interpretable Emotional Dialogue Agents via Chain-of-Thought Reasoning</b>   2025 Spring                                                                                                                                                                                                                                                                                                    |  |
|             | <ul style="list-style-type: none"> <li>Developed a unified framework for machine emotional intelligence that integrates emotion recognition, cause inference, shift detection, and dialogue generation using Chain-of-Thought (CoT) reasoning.</li> </ul>                                                                                                                                               |  |
|             | <b>ICU Simulation and Resource Optimization</b>   2024 Fall                                                                                                                                                                                                                                                                                                                                             |  |
|             | <ul style="list-style-type: none"> <li>Utilize simulation frameworks to optimize resources in the Intensive Care Unit (ICU), including critical components such as beds and nursing staff.</li> <li>Key components include ICU queue simulation, nursing staff workflow simulation and various optimization methods such as exhaustive search, heuristics, and Pareto frontier-based method.</li> </ul> |  |
| INTERNSHIPS | <b>Shenzhen Guanghuiyuan Asset Management Co., Ltd.</b>   Shenzhen, China 2023.07 - 2023.08                                                                                                                                                                                                                                                                                                             |  |
|             | <ul style="list-style-type: none"> <li>Studied and participated in value investment research.</li> </ul>                                                                                                                                                                                                                                                                                                |  |
| SKILLS      | <b>Languages:</b> Mandarin, English.<br><b>Programming:</b> Python, TensorFlow, PyTorch, C, C++, Linux.<br><b>Others:</b> Solid math foundation. Eager to learn new things.                                                                                                                                                                                                                             |  |

Last updated: 2026.05.14